

Planning Logic

Project Name:

Voice-Based Concept Understanding Analyser (VBCUA)

Objective

The project aims to evaluate a student's conceptual understanding through voice input by combining speech recognition, semantic similarity analysis, fluency analysis, and AI-generated feedback.

Project Planning

Phase 1 – Requirement Gathering

- Identified the project objective.
- Collected functional and non-functional requirements.
- Finalized the technology stack.

Phase 2 – System Design

- Designed the project workflow.
- Prepared solution architecture.
- Created project documentation.

Phase 3 – Development

- Implemented the Streamlit frontend.
- Integrated OpenAI Whisper for speech-to-text conversion.
- Added semantic similarity analysis using Sentence Transformers.
- Developed fluency analysis.
- Implemented AI feedback generation.

- Generated downloadable PDF reports.

Phase 4 – Testing

- Tested individual modules.
- Verified transcript generation.
- Validated semantic scoring.
- Checked PDF report generation.
- Performed user acceptance testing.

Phase 5 – Deployment

- Hosted the application using Streamlit Cloud.
- Uploaded the source code to GitHub.
- Prepared project documentation and demonstration.

Milestones

Milestone	Status
Requirement Analysis	Completed
Design	Completed
Development	Completed
Testing	Completed
Deployment	Completed